

ing unavailability of AC mains power, the battery kicks in to run the inverter.

Benefits

Keeps your devices safe. An online UPS system completely isolates the connected devices from the source by placing itself directly in the power path. The main power route being the rectifier-inverter path, disproportionate DC current and voltages are rectified, removing all sorts of voltage inconsistencies, and thus providing a uniform smooth output to the connected device. This saves sensitive high-end equipment from possible damage.

Eliminates waiting time. Since the battery and the inverter are both connected to the main power path of the device, when input current cuts off during power failure, the battery immediately powers up the inverter, keeping the system running without any pause whatsoever. No transfer of power path is required, essentially eliminating the waiting time for the backup to kick in. Mission-critical activities and data centres, where down-time elimination is very essential, stand to benefit greatly from this feature.

Long lifetime and easy installation. Online UPS systems, if used under suitable conditions, can go a long way. Batteries may need to be replaced every three to four years, but rest of the device remains functional for a long time. Also, the device installation is simple, requiring only the connection of power grid and devices to the UPS.

Drawbacks

High price. Online UPS cost more than double the line-interactive UPS, that too at the basic level. While proper usage can avoid unnecessary repair costs, the initial cost of investment is quite high, making many medium-size businesses prefer other UPS systems.

Why delta-conversion UPS systems are not popular

Because of the extra transformer and other components added to the architecture, delta-conversion UPS systems cost 30 to 40 per cent more than their double-conversion counterparts. Consumers in India are not willing to spend extra. To survive in this cost-conscious competitive market, manufacturers produce only a limited stock of such systems. However, big companies and multinationals, who are well informed, go for delta-conversion models.

—Sachin Mittal, branch manager, SPOC Power Solutions

B.N. Mohapatra, purchase manager at DHN Technologies, shares, "Online UPS are most suited for critical applications like healthcare, data centres and any other sector where continuous power supply is most important." An entry-level 3kVA online UPS can cost anywhere between ₹ 20,000 and ₹ 30,000. Higher-rating 60kVA systems cost up to ₹ 600,000, while mission-critical UPS (such as the ones used in data centres) cost even more.

Loss of efficiency in double-conversion. As stated before, the two-step AC-DC conversion process in online double-conversion systems causes significant loss of energy, lowering the system efficiency down to 89 per cent. Delta-conversion online architecture solves this problem. Unfortunately, there is not much awareness about this benefit of delta-conversion UPS.

Nevertheless, industry players are hopeful that the market for delta-conversion UPS will pick up in the future. Kunwer Sachdev, managing director of Su-Kam, explains, "Delta-conversion online UPS are available in the Indian market. As all motor applications in manufacturing units use delta-conversion online UPS, the market for their application is huge."

What's the ROI?

Annual servicing is a major operational cost, accounting for almost 12 to 15 per cent of the original price of the UPS. Therefore it is important to check service centres and after-sales policy of the UPS manufacturer.

An online UPS system's return

on investment (ROI) is based on the amount of load used, application and purpose, downtime and efficiency of the UPS.

Rajesh Manocha, owner of Aropower Technologies, says, "An online UPS can start giving ROI after one or two years if there is a considerable use of the UPS with no downtime. It will take longer to realise the ROI if the UPS suffers a large downtime. Since the UPS architecture can be customised to users' requirements, the chances of downtime can also be controlled likewise. But the cost of the UPS will vary accordingly."

Any equipment can start giving ROI if the firm is able to utilise its full potential. Mittal explains, "Say, every day there is a cumulative power cut of 30 minutes. The work saved in those 30 minutes, which could have been lost without the UPS, becomes the ROI. For a BPO, for example, saving 30 minutes worth of time and work can save a huge amount of data and finance for the company."

Solar is the future

In the future, customers can expect higher-efficiency online UPS along with lower maintenance costs. "The UPS technology is gaining steam in the Indian market and solar is the future with respect to power backup industry. However, there are many places that have a shortage of installation space and hence the people in those areas are bound to purchase normal online UPS," says Kartik Sachdev, marketing head, Solar, Su-Kam. **EFY**